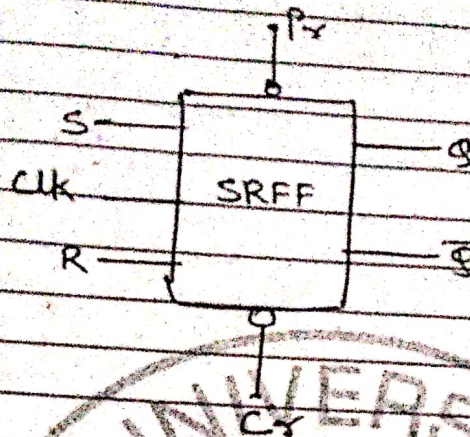


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Question No.

This SRFF is symbolically shown as



I/P			O/P	
ck	cr	pr	Q _{n+1}	
1	1	1	Q _{n+1}	Normal
0	0	1	0	clear
0	1	0	1	preset

In symbol, P_r & C_r are shown active low.
I/p's

Bcz FF is cleared i.e. $Q=0$ or FF is preset i.e. $Q=1$, only if signal applied to P_r or C_r is low.

These P_r & C_r I/p are also called as async I/p bcz they can be applied to the ckt. at any time betⁿ clk pulses & will give effect on o/p. i.e. they are not synchronized with clk.

This SRFF can be used in control applⁿ where we may think in terms of setting flag wth some condⁿ & resetting flag when condition is changed.

Dis. adv. of SRFF →

- ① In this $S=R=1$, is not allowed state or forbidden state.
- ② Also to set FF it req. $S=1$ & to reset FF it req. $R=1$. So i.e. 2 diff. signals are req. for set/reset, which is not desirable in some applⁿ.

5 10 10 1 6 10 10 1

SRFF w/f.

clk.

S

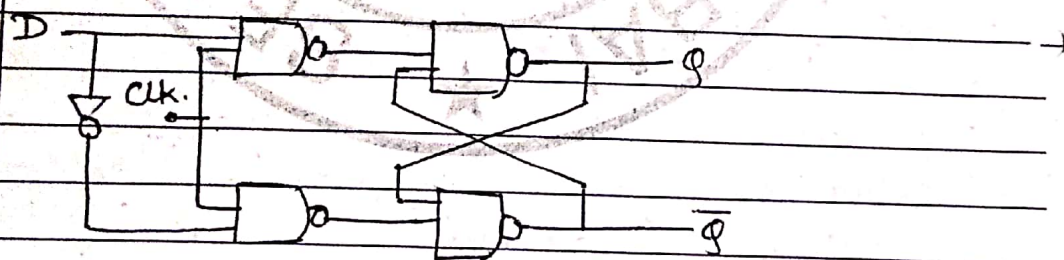
R

Q

ignored as $clk=0$.D-FF $\rightarrow$ Delay FF.

Is FF having only one i/p, referred as D i/p or data i/p.

Ckt. is as below:



As shown, it is obtained from S-R FF, by connecting an inverter betⁿ two i/p's of SRFF.

Here when $D=0$, i.e. $S=0, R=1$ o/p $Q=0$.

$D=1$, i.e. $S=1, R=0$, o/p $Q=1$.

$\therefore$ TT will be

D	Q_{n+1}
0	0
1	1

Q_n Q_{n+1}
 0 0 0
 1 0 1

$$Q_{n+1} = D\bar{Q}_n + \bar{D}Q_n$$

$Q_{n+1} = D$ charac. eqⁿ G13

Here o/p is same as data i/p. i.e o/p at the end of clk pulse is same as i/p ($D_{\text{before clk}}$). That means i/p data appears at o/p, after some delay (of 1 clk pulse.)

As data is transferred from i/p to o/p is delayed, name is Delay FF. It is widely used in computer sys., simply for storing a bits of info.

Symbol :

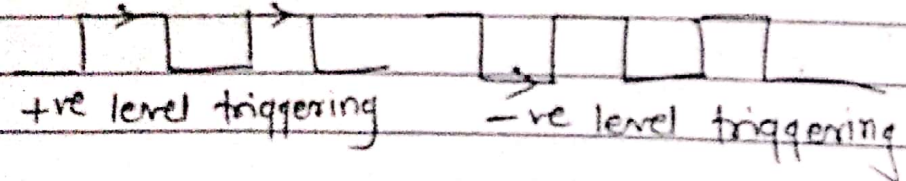


Before w/f Lets discuss about clk.



clk has 2 levels low & high. If FF responds to high level of clk, means o/p of FF changes w/o i/p only when clk = high level 1. This is called as high or +ve level triggering.

If FF responds to low level of clk then -ve or low level triggering.

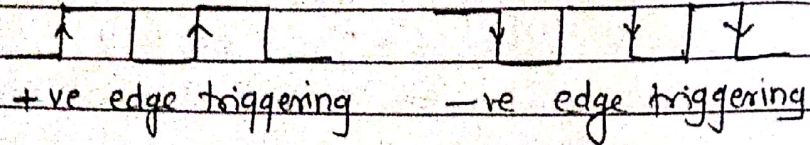


This was level triggering.

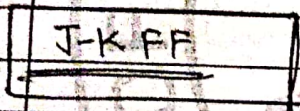
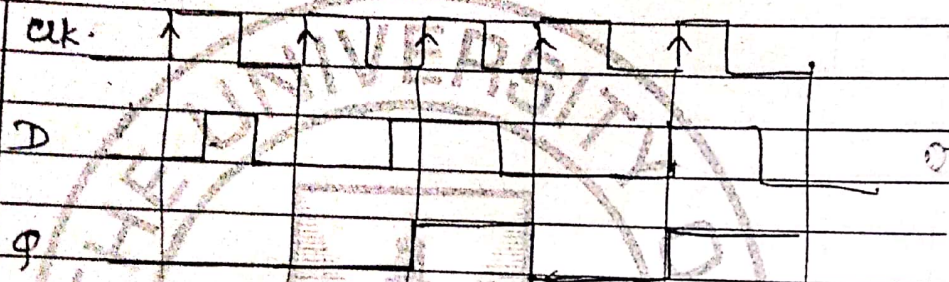
You can have another type of triggering called as edge triggering.

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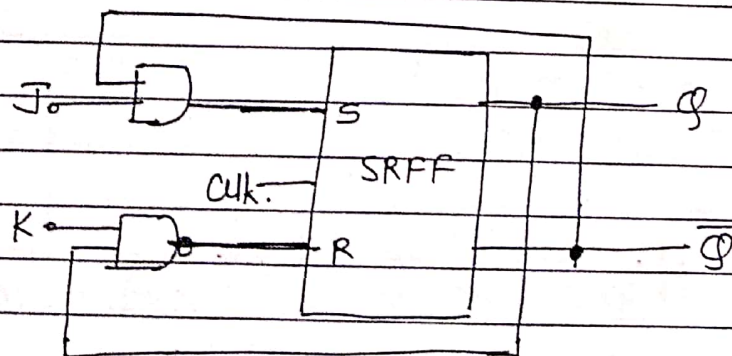
When FF responds to its i/p only at rising/+ve & falling/-ve edges.



Let's see w/f of D-FF for +ve edge triggering.



SRFF's $S=R=1$ condⁿ/forbidden state can be eliminated by converting it into a JK-FF. This JKFF is modificalⁿ to SRFF to overcome its drawback. as shown



As shown in the ckt. diag. the data i/p's are J & K. And these J & K i/p's are ANDed with Q & $\bar{Q}$ respec. to obtain

S & R inputs

Such that now

$$S = J\bar{Q}$$

$$R = KQ$$

To understand its operation let's proceed with TT.

Data = J, K		Previous o/p		J, K to S, R		Output
J	K	Q_n	$\bar{Q}_n$	S	R	Q_{n+1}
0	0	0	1	0	0	$Q_n = 0 \rightarrow 0$
0	0	1	0	0	0	$Q_n = 1 \rightarrow 1$
0	1	0	1	0	0	0
0	1	1	0	0	1	0
1	0	0	1	1	0	1
1	0	1	0	0	0	1
1	1	0	1	1	0	0
1	1	1	0	0	1	0

Consider 1st case:

$J=K=0$, and previous $Q_n=0, \bar{Q}_n=1$ With this new S & R i/p's becomes $S=0, R=0$ when $S=R=0$, it follows previous o/p $\therefore$

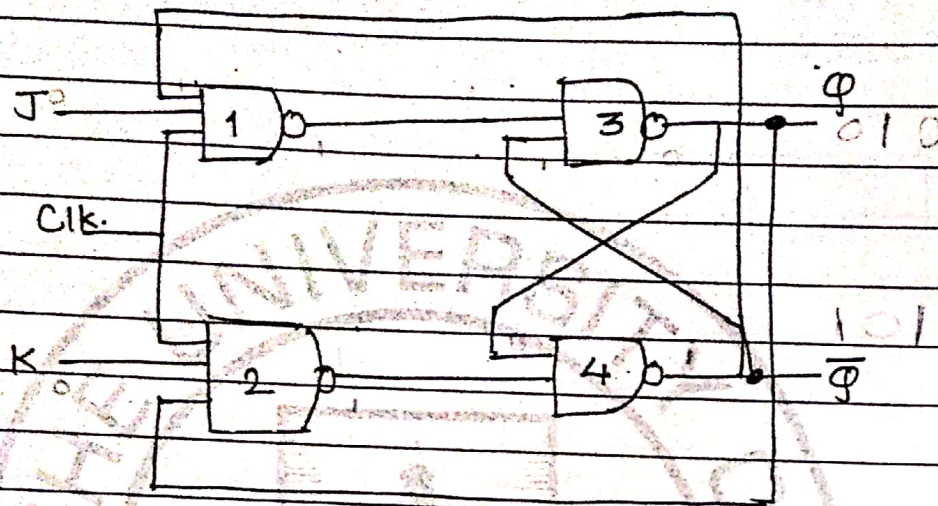
$$Q_{n+1} = Q_n = 0$$

similarly for all other combinations of $J \neq K$ along with both the state of o/p has been explained in TT.

This can be reduced further as :-

J	K	Q_{n+1}
0	0	Q_n
0	1	0
1	0	1
1	1	$\bar{Q}_n$

Now instead of using SRFF with AND, one can design JKFF using only NAND gates simply by adding an extra i/p terminal to i/p gates as shown.



Here when

→ $J=K=0$, o/p of gate 1 & 2 is high. Now if $\phi=0$, $\bar{\phi}=1$ then o/p of 4 = 1, $\therefore$ o/p of 3 = 0. It remains in same state.

If $\phi=1$ & $\bar{\phi}=0$, then o/p of 3 = 1. Makes o/p of 4 = 0. i.e. with $J=K=0$, o/p remains same as previous o/p.

$$\phi_{n+1} = \phi_n$$

→ $J=1$, $K=0$. o/p of 1 & 2 $\phi_n=0$, $\bar{\phi}_n=1$ then o/p of 1 = 0, 2 = 1. Now 1 makes 3 = 1. And o/p of 4 = 0. i.e. now $\phi=1$ $\bar{\phi}=0$.

Now if $\phi_n=1$ & $\bar{\phi}_n=0$ then o/p of 1 & 2 = 1, o/p of 2 = 1. Force o/p of 3 = 1. Now o/p of 4 = 0. i.e. $\phi=1$, $\bar{\phi}=0$

$\therefore J=1$, $K=0$ always gives $\phi=1$, $\bar{\phi}=0$.

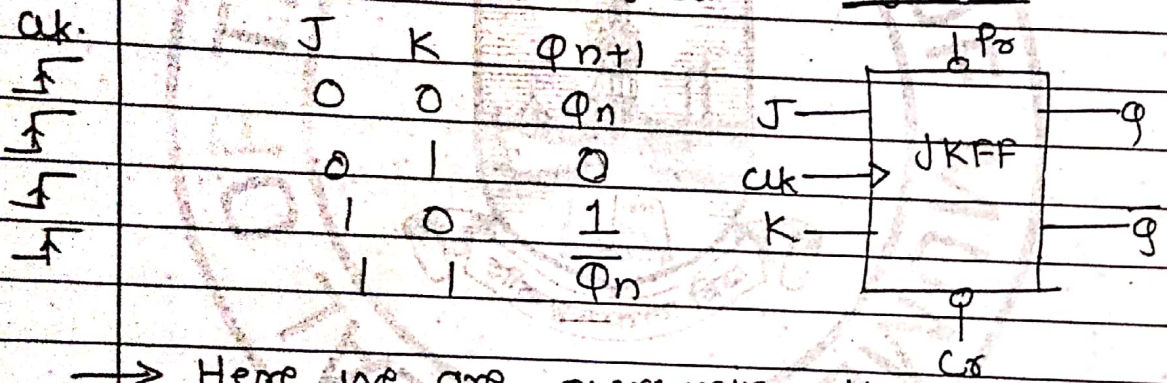
→ $J=0$, $K=1$, with $\phi=0$, $\bar{\phi}=1$. o/p of 1 = 1, 2 = 1. o/p of 3 = 0, $\therefore$ 4 = 1 i.e. $\phi=0$, $\bar{\phi}=1$.

if $\phi=1$ & $\bar{\phi}=0$, then 1 = 1, 2 = 0, forces o/p of 4 = 1 i.e. $\bar{\phi}=1$ and $\phi=0$. i.e. $J=0$, $K=1$, $\phi=0$, $\bar{\phi}=1$.

→ $J=K=1$. consider $\phi=0, \bar{\phi}=1$.
 o/p of 1 = 0, 2 = 1. 1 forces 3 = 1.
 ∴ o/p of 4 = 0. ie $\phi=1, \bar{\phi}=0$.
 ie we gen $\phi_{n+1} = \bar{\phi}_n$ considered.

Similarly for $J=K=1$. & $\phi=1, \bar{\phi}=0$.
 o/p of 1 = 1, 2 = 0, 2 force 4 = 1.
 Moment 4 = 1, o/p of 3 = 0 ie $\phi=0, \bar{\phi}=1$.
 Here also $\phi_{n+1} = \bar{\phi}_n$

→ ∴ TT is verified. Symbol



→ Here we are overcoming the drawback of SRFF by using F/B connection from o/p to the i/p gates. In this case we have assumed the FF is edge triggered. ie our ckt. is triggered for +ve edges of clk. So TT is getting verified for these edges.

Now see last row of TT, $J=K=1$, & $\phi = \bar{\phi}_n$ ie at the arrival of each clk., FF will keep on changing state.

$J=K=1$ 1st clk. $\phi=0, \bar{\phi}=1$.
 2nd clk. $=1, =0$
 3rd clk. $=0, =1$.

So wost clk received, states keeps on changing called as Toggle. means o/p toggles.

Question No.

wlf will be

clk

Q_{n+1}

$$Q_{n+1} = \overline{Q_n}$$

Now instead of edge triggered FF, consider that it is level triggered.

Δt



t_p

When level trigger, FF will respond to entire period t_p of clk.

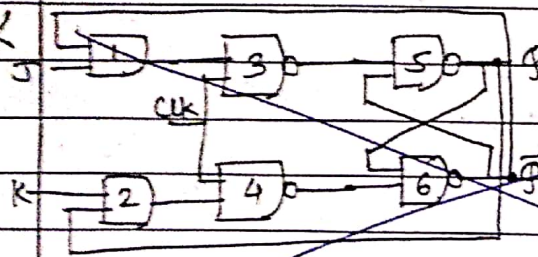
Now here, if we consider $J=K=1$ & $Q=0$ then after some time say Δt $Q=1$.

Now state is $J=K=1$ & $Q=1$.

After next period Δt $Q=0$. This continues for duration t_p . O/p is continuously changing betⁿ 0 & 1. And at the end of clk pulse, value of Q is uncertain.

This situation is called as Race around condⁿ.

This prob. in JKFF, when it is level triggered is overcome by Master-slave JKFF.



$$J=K=1, Q=0, \overline{Q}=1$$

$$O/p 1=1, 2=0, 3=0, 4=1,$$

$$5=1, 6=0. \therefore Q=1, \overline{Q}=0$$

$$J=K=1, Q=1, \overline{Q}=0$$

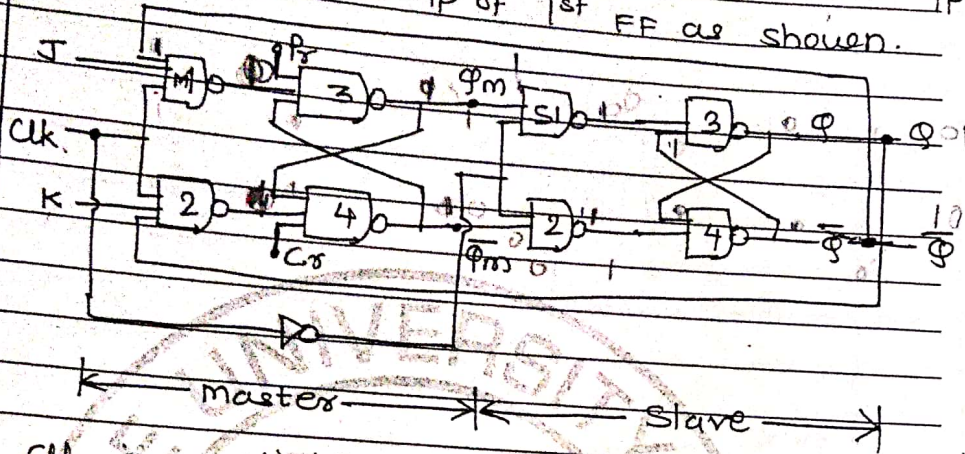
$$O/p 1=0, 2=1, 3=1, 4=0,$$

$$5=1, \therefore 5=0. Q=0, \overline{Q}=1$$

$$Q_{n+1} = \overline{Q_n}$$

Master Slave JKFF

It is cascade of 2 SRFF, with F/B from o/p of second FF to the i/p of 1st FF as shown.



Ck is applied to Ck has 2 SRFF's. J & K inputs are 1st FF i/p's. while their o/p is applied as i/p to next FF.

Ck is applied to 1st stage and inverted Ck is applied to 2nd stage. When Ck = 1, 1st FF is enabled and o/p Q_m & $\bar{Q}_m$ responds to inputs J & K (as per JKFF TT.)

At that time, 2nd FF is inactive bcz Ck = low

Now when Ck is low, 1st FF is inactive and 2nd FF is enabled, as its Ck = high. So o/p Q & $\bar{Q}$ follows its i/p Q_m & $\bar{Q}_m$ respect. (2nd & 3rd row of TT.)

→ Consider, $J=K=1$, $Q=0$, $\bar{Q}=1$, Ck = 1.

O/p $m_1=0$, $m_2=1$, $m_3=1$, $m_4=0$. $Q_m=1$, $\bar{Q}_m=0$

O/p of $S_1=S_2=1$ as Ck = 0, $\therefore Q=0$, $\bar{Q}=1$.

Now $J=K=1$, $Q=0$, $\bar{Q}=1$, Ck = 0.

O/p $m_1=m_2=0$, $\therefore Q_m$ & $\bar{Q}_m$ remains unchanged.
ie $Q_m=1$, $\bar{Q}_m=0$.

O/p of $S_1=0$, $S_2=1$, $\therefore S_3=1$ & $S_4=0$.

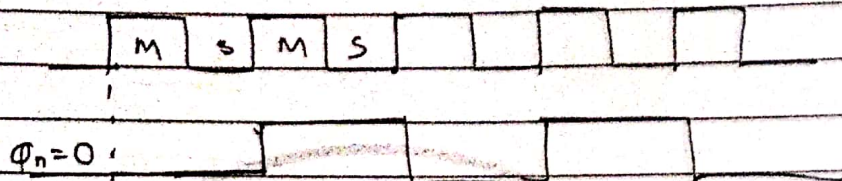
ie now $Q=1$ & $\bar{Q}=0$.

At the end of Ck pulse we get $Q_{n+1} = \bar{Q}_n$

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So race around condⁿ is overcome.

Here o/p Q & $\bar{Q}$ are same as Q_m & $\bar{Q}_m$.
ie 2nd stage simply follows 1st
state. That's why called as slave. And
1st as Master.

~~JK~~

JKFF

J	K	Q_n	Q_{n+1}	Q_n	JK	00	01	11	10
0	0	0	0	0	0			1	1
0	0	1	1	1	1	1	1		
0	1	0	0						
0	1	1	0						
1	0	0	1						
1	0	1	1						
1	1	0	1						
1	1	1	0						

$$Q_{n+1} = J\bar{Q}_n + \bar{K}Q_n$$

TT =	J	K	Q_{n+1}
	0	0	Q_n
	0	1	0
	1	0	1
	1	1	$\bar{Q}_n$

Question No.

T-FF (Toggle FF)

In JKFF, if $J=K$, then resultant FF is called toggle / T FF.
It has only one ilp, T - ilp.

JKFF

Inputs: T , clk , $\bar{Q}$

Output: Q

T	Q_n	Q_{n+1}
0	0	0
0	1	1
1	0	1
1	1	0

$Q_{n+1} = T \oplus Q_n$

$$Q_{n+1} = T \oplus Q_n$$

When $T=0$, ie $J=K=0$, $Q_{n+1} = Q_n$
 $T=1$, ie $J=K=1$, $Q_{n+1} = \overline{Q_n}$

T	Q_{n+1}
0	Q_n
1	$\overline{Q_n}$

When $T=1$, o/p keeps on changing, at the arrival of each clk. See here, that JKFF is edge triggered.

One can't get T-FF from SRFF, bcz $S=R=1$ is not allowed state in SRFF.

FF Conversions $\rightarrow$

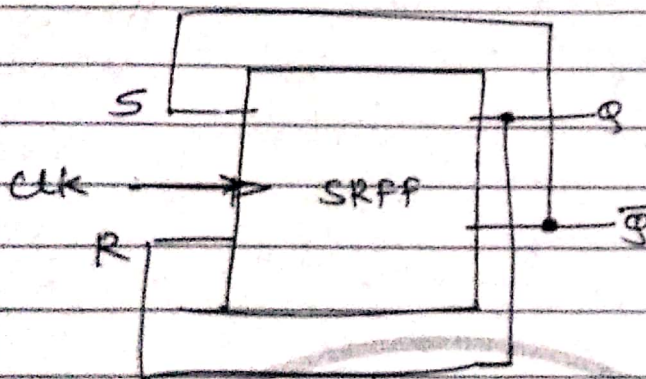
SRFF to JKFF by adding AND gates.

SRFF to D by an INV betⁿ S & R.

JKFF to T by shorting J & K inputs.

JKFF to D by an INV betⁿ J & K.

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No.SRFF as toggle switch

clk acts as toggle switch, Q changes at each clk pulse.